Multi-Mode Transfer Matrix Method (MMTMM) for Leaky Waveguide Dispersion characterization.

Description: This document summarizes the application of the MMTMM for characterizing the Bloch dispersion of leaky wave antennas (LWAs). MMTMM is a relevant novel method of determining the Bloch dispersion of complex leaky wave antenna structures with the aid of a full-wave electromagnetics solver. The effectiveness of this method is found in its compatibility with almost all commercial full wave solvers and its algorithmic simplicity.

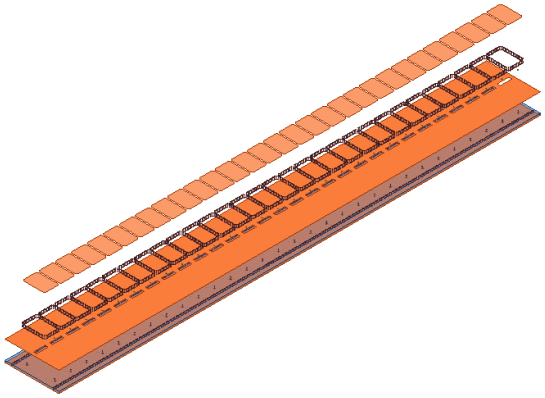


Figure 1: Expanded view of an LWA.

To start, MMTMM is applied to a leaky wave antenna with transverse slots using Ansys HFSS. [1][2] have already developed a set of transcendental equations which have been shown to correctly describe the dispersion characteristics of the aforementioned leaky wave structure. The unit cell model is excited with modal waveguide ports which each contain 4 modes at each end of the waveguide. The port height is extended as per [3] for stronger solution convergence. Finite conductivity boundaries are applied to the side and vertical walls of the simplified substrate integrated waveguide (SIW) for accurate modelling of the conductor loss. The S matrix data from the solved structure depicted below is extracted and subjected to MMTMM based post processing.

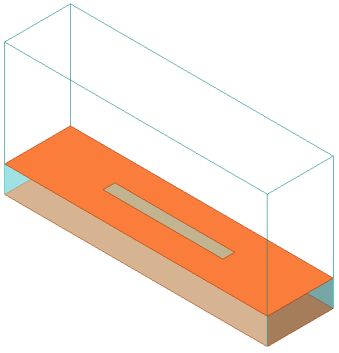


Figure 2: MMTMM unit cell model of LWA with transverse slots (LWA-TS).

A particle swarm optimization algorithm developed in MATLAB is used to numerically compute the wavenumber of the SIW geometry based on the transcendental equation presented in [2], in which the authors determine the solution to the leaky wave number of the studied LWA geometry by imposing magnetic field boundary integrals on the slot geometry. The extracted dispersion characteristics from the transcendental equation in [2] and the MMTMM method presented in [3] are shown below.

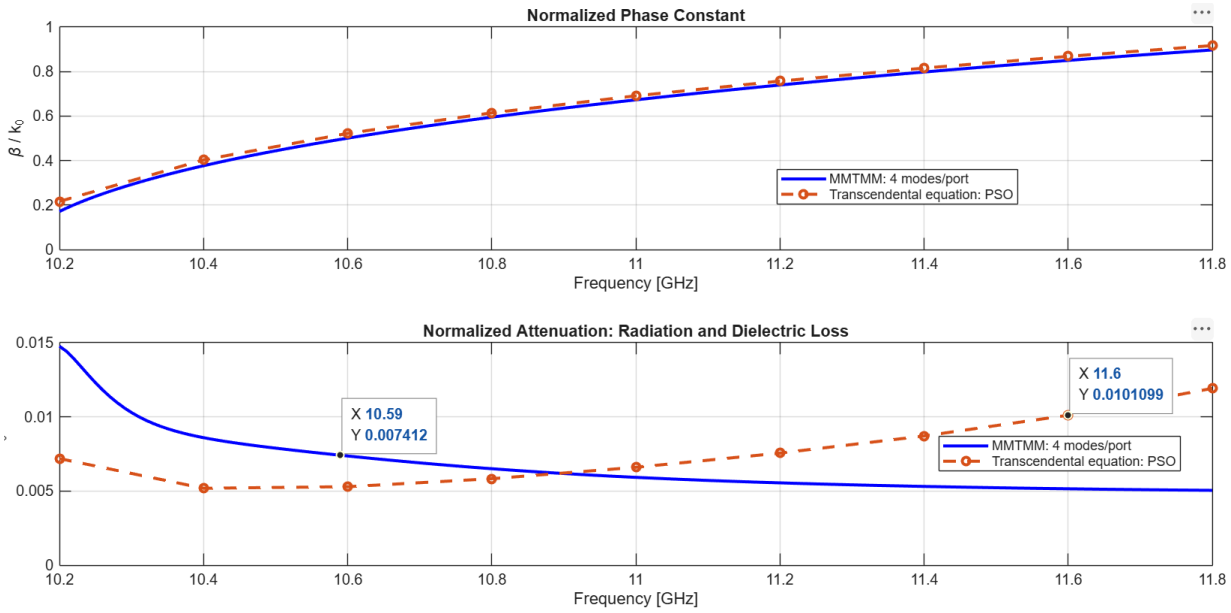


Figure 3: MMTMM and magnetic field boundary integral equation extracted Bloch dispersion.

The dispersion curves shown above quantify the Bloch phase and leakage (conductor loss + dielectric loss + radiation loss) as a function of frequency from 10.2 to 11.8 GHz. While the Bloch phase extracted from both methods matches almost perfectly, there is a fairly strong difference in the extracted leakage rates.

Alternatively, the ports are positioned at the vertical planes of each slot as done in [3] for the MMTMM Bloch mode extraction of a *longitudinal* slot LWA. This port setup better captures the periodicity of the structure and therefore provides better determination of the improper (radiating) leaky mode propagating through the structure. The updated unit cell structure is visualized below.

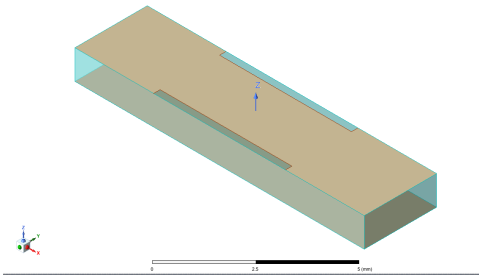


Figure 4: MMTMM unit cell of transverse slotted LWA.

Additionally, prior to MMTMM dispersion characterization, the number of modes was increased to 8. To ensure proper port polarity between all modes, a “port only” solution was run to visually inspect the port field distribution and assign integration lines along the dominant fields.

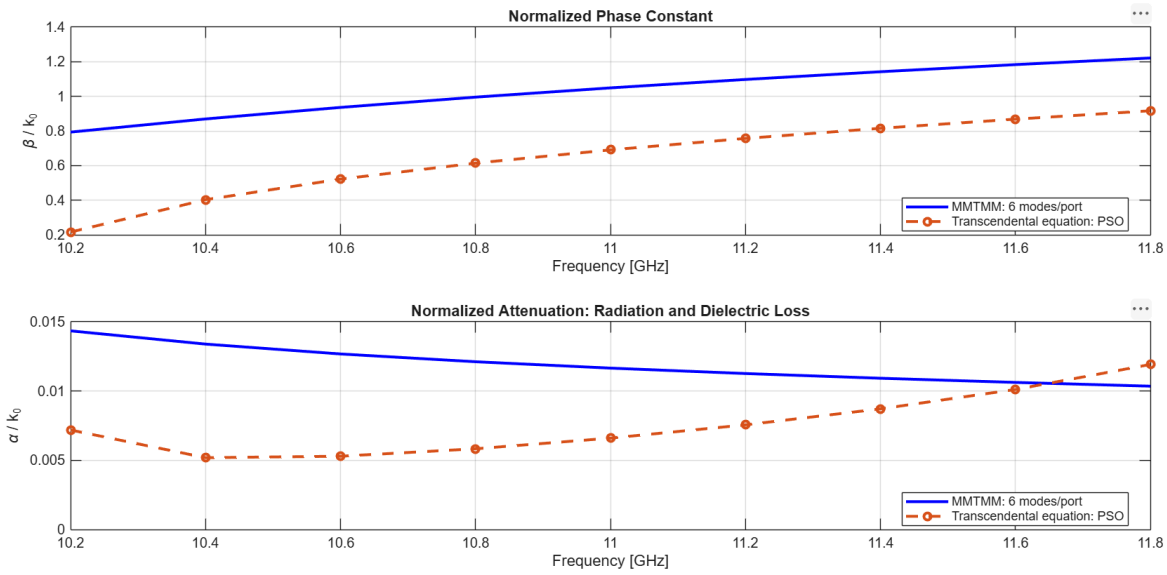


Figure 5: Initial comparison of MMTMM and numerical results for transverse slotted LWA.

Stronger disagreement is produced by the offset unit cell simulation. Inspection of the MMTMM simulation results seems t

References

[1] J. Liu, D. R. Jackson, and Y. Long, “Modal analysis of dielectric-filled rectangular waveguide with transverse slots,” *IEEE Transactions on Antennas and Propagation*, vol. 59, no. 9, pp. 3194–3203, Sept. 2011, doi: 10.1109/TAP.2011.2161444.

[2] J. Liu, D. R. Jackson, and Y. Long, “Substrate integrated waveguide (SIW) leaky-wave antenna with transverse slots,” *IEEE Transactions on Antennas and Propagation*, vol. 60, no. 1, pp. 20–29, Jan. 2012, doi: 10.1109/TAP.2011.2167910.

[3] S. Garcia-Martinez, F. Giusti, F. Mesa, A. Tamayo-Dominguez, and O. Quevedo-Teruel, “Accurate full-wave Bloch analysis of general 1D-periodic open structures,” *IEEE Transactions on Antennas and Propagation*, vol. 73, no. 11, pp. 8769–8778, Nov. 2025, doi: 10.1109/TAP.2025.3587901.